



REPUBLIC OF THE PHILIPPINES

POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

COLLEGE OF COMPUTER AND INFORMATION SCIENCES



CERTIFICATION OF VALIDATION

Date: October 11, 2025

To Whom It May Concern,

This is to formally certify that I, JC Albert C Peralta, a Geospatial Machine Learning Consultant, have reviewed and validated the dataset utilized in the study entitled: **“SETTLENET: CONVNEXT ENCODER-DECODER SEMANTIC SEGMENTATION OF IRREGULAR HIGH-DENSITY SETTLEMENTS WITH MULTIMODAL FUSION OF RGB SATELLITE, BUILDING COUNT, AND HEIGHT MAPS”**.

I have completed a thorough validation of the study's datasets and methodology, and I confirm the consistency and integrity of the satellite imagery, building density, and building height datasets used, as well as approve of the choice of encoder-decoder architecture to accomplish the segmentation task.

I present the following findings and ethical concerns as guidance of the researchers and their reviewers.

- 1.) **Data Licenses.** I have reviewed and verified the licenses of the datasets. The use of Maxar Open Data (CC BY-NC 4.0) and Google Open Buildings 2.5D (CC BY 4.0/ODbL) is fully compliant with their respective licenses and appropriate for this research.
- 2.) **Responsible Dissemination of Research Output and the "Do No Harm" Principle.** While the intention of the study is undoubtedly to provide data for positive interventions, the generated binary masks and associated geolocational data pinpointing exact locations of informal settlements, especially emerging ones, are highly sensitive. If made publicly and easily accessible, this data could be weaponized by actors with interests not aligned with the well-being of these communities. To protect this vulnerable sector, I strongly recommend that the final output, particularly the segmentation masks, must NOT be made publicly accessible. However, if there are plans to share this data with legitimate stakeholders, it is imperative that you coordinate closely with the Quezon City local government and, if possible, with representatives from the informal communities themselves.

Should further verification be required, I can be reached via email at jcacperalta@gmail.com.

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